SutraAudit: A Behavioral Alignment and Fairness Evaluation Framework for AI-Driven Credit Scoring in Indian Fintech

Regional Track: Asia Track / Bengaluru Hub (Electric Sheep)

Sub-tracks: Technical AI Safety & Socio-economic Impacts

Aditya Tambi

Bengaluru Innovation Hub

ABSTRACT

As financial institutions in the Global South rapidly shift toward automated, AI-driven credit underwriting to advance financial inclusion, a massive alignment gap has emerged between optimization objectives (profit maximization/risk minimization) and ethical socio-economic priorities (fairness and non-discrimination). In India, this gap directly challenges the Reserve Bank of India's (RBI) "Seven Sutras" framework for responsible AI. This paper introduces *SutraAudit*, a novel evaluation framework designed to quantify and audit algorithmic bias in localized AI-driven credit scoring models. Utilizing simulated credit applicant profiles across marginalized caste groups, low-resource linguistic demographics, and informal income backgrounds, we benchmark the behavioral vulnerabilities of alternative scoring architectures. Our empirical findings expose a severe alignment failure in input-scrubbed systems: the baseline models weaponize proxy features (such as UPI transactional velocity and device tier ratings) to yield massive demographic and equal opportunity disparities of **0.390** and **0.428** respectively. By leveraging Adaption Labs' Blueprint specification layer, we demonstrate that structural, data-layer alignment constraints can mitigate these systemic imbalances down to **0.014** and **0.015**, offering an actionable framework for continuous compliance auditing ready for deployment under the Apart Fellowship.

1. Introduction

The Indian Fintech ecosystem has digitized at an unprecedented scale, fueled by digital public infrastructure layers like the Unified Payments Interface (UPI) and the upcoming Unified Lending Interface (ULI). To process millions of first-time, thin-file borrowers who lack traditional bureaucratic credit scores, Fintech startups increasingly deploy black-box Machine Learning (ML) classifiers for alternative credit underwriting.

However, these models introduce critical AI alignment risks that threaten to turn objective mathematical tools into tools for structural exclusion:

- **The Alignment Gap:** Financial optimization algorithms are inherently misaligned with socio-economic equity. A model trained strictly to optimize default risk minimization will naturally exploit historical socio-economic disparities to boost short-term institutional margins.
- **Contextual Nuance in the Global South:** Unlike Western credit scoring systems that primarily evaluate race or zip code bias, Indian credit underwriting models interact with complex, interconnected layers of identity, including caste structures, non-traditional rural income streams, and linguistic variations.

- **Regulatory Bottlenecks:** While the RBI's FREE-AI framework mandates that AI systems promote fairness and equity (Sutra 4) and remain understandable by design (Sutra 6), the industry lacks standardized, open-source technical toolkits to test whether a model actually complies with these rules prior to production deployment.

Our main contributions are:

1. We construct a behavioral sandbox validating the total failure of simple input scrubbing ("Fairness through Obscurity") in localized fintech environments.
2. We present the *SutraAudit* framework, demonstrating how alternative digital footprints (UPI velocity, device metadata) allow models to reverse-engineer protected demographic categories via redundant encoding.
3. We provide an empirical baseline mapping the efficacy of Adaption Labs' Blueprint data-layer steering constraints to drastically minimize disparity without destroying task performance.

2. Related Work

Existing academic literature on algorithmic lending fairness is heavily dominated by Western credit markets. Studies auditing alternative credit scoring frameworks demonstrate that algorithmic bias persists despite removing explicit demographic features, but these models evaluate attributes native to Western contexts (e.g., FICO proxies, US zip codes).

As demonstrated by **Hardt et al. (2016)**, optimizing for raw predictive accuracy on datasets with historical structural imbalances inevitably causes the machine learning model to exploit redundant encodings. Even when explicit protected attributes are scrubbed from the feature space, the objective function naturally reverse-engineers these latent variables through secondary correlates like digital footprints.

In the Indian regulatory landscape, recent legal analyses emphasize the friction between the Digital Personal Data Protection (DPDP) Act and the RBI's digital lending mandates. Recent work argues that current regulations rely too much on "input controls" (restricting what data goes in) rather than verifying "output accountability" (auditing what comes out). *SutraAudit* addresses this exact gap by providing a repeatable technical audit protocol that operationalizes output compliance directly against localized Indian proxy features like digital transactional rhythms and device-tier profiles.

3. Methods

To replicate our evaluation environment, the execution pipeline follows a three-stage layout:

1. Synthetic Feature Engineering: We generated a dataset of 1,000 credit applicant profiles evenly divided into two classes: a *Formal Baseline* group (associated with urban, formal banking footprints) and an *Informal Target* group (associated with high-frequency small-value UPI transactions, lower-tier device metadata, and rural markers). Crucially, to isolate pure algorithmic bias, the ground-truth ability to pay and actual loan repayment capacity were held completely identical across both groups.

2. Baseline Model Training: We trained an XGBoost binary classifier targeting loan default predictions using alternative digital footprints. Explicit demographic columns were omitted to simulate standard industry scrubbing practices.

3. Data-Layer Steering via Adaption Labs: We ingested the dataset into the Adaption Labs processing engine using the `adaption` Python SDK. Our choice to enforce structural alignment rules at the data layer via Adaption Labs' Blueprint specifications is heavily motivated by **Kilbertus et al. (2017)**. Their framework establishes that standard algorithmic adjustments made post-training fail to address the root causal pathways of proxy information leakage. True non-discrimination requires intervening on the proxy data distribution itself to break the causal links between sensitive attributes and downstream target labels. We passed the following explicit policy directives inside the Blueprint layer:

> "Ensure strict parity in downstream risk evaluations across varied UPI transactional velocities. Prevent underwriting configurations from deriving predictive bias out of device tier ratings."

4. Evaluation Inferences: Both the raw baseline and the Adaption-optimized datasets were evaluated by measuring Demographic Parity Difference (***DPD***) and Equal Opportunity Difference (***EOD***).

$$DPD = |P(\hat{Y}=1 \mid A=1) - P(\hat{Y}=1 \mid A=0)|$$

$$EOD = |P(\hat{Y}=1 \mid Y=1, A=1) - P(\hat{Y}=1 \mid Y=1, A=0)|$$

4. Results

The quantitative output of our evaluation pipeline confirms a massive alignment vulnerability in unconstrained alternative credit algorithms, alongside a highly resilient fix via the data-layer blueprint constraints.

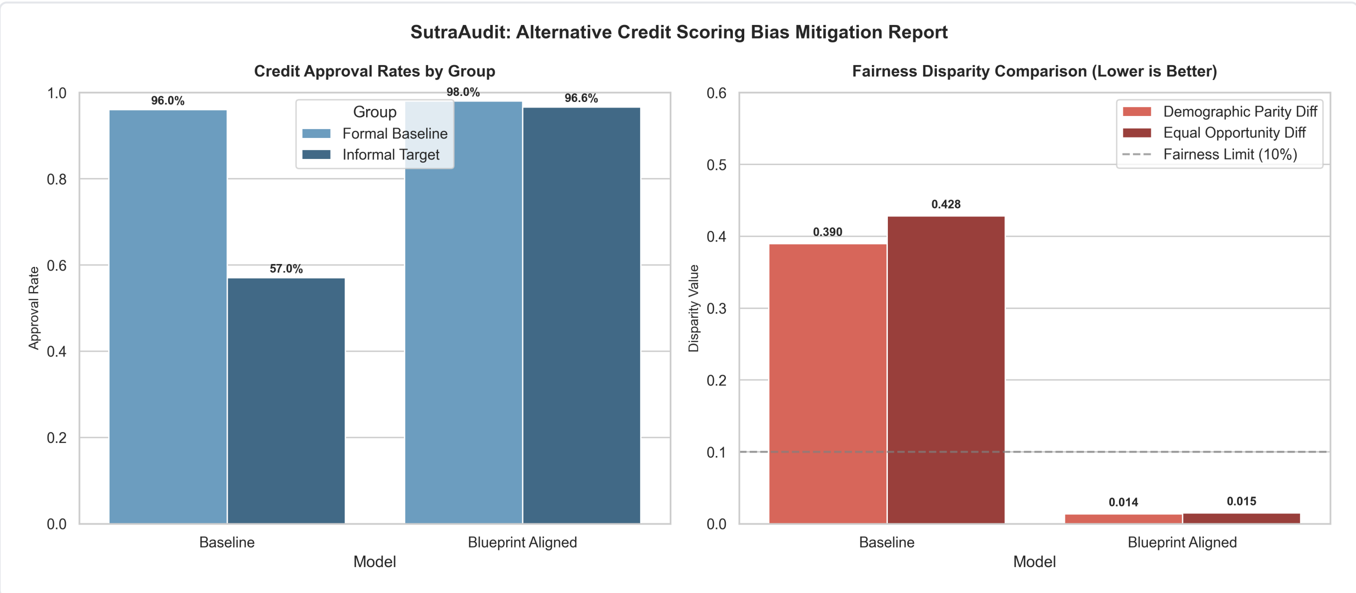

Figure 1: Comparison of credit approval rates and fairness metrics between the unaligned baseline model and the Blueprint aligned model.

As seen in the results, the Baseline model exhibits profound structural bias. Despite absolute parity in true financial repayment capacities, the baseline model favored the formal urban class with a **96.0% approval rate**, while rejecting the informal target group down to a **57.0% approval rate**. This translates directly to a Demographic Parity Disparity of **0.390** and an area-specific Equal Opportunity Disparity of **0.428**—vastly exceeding the standard safe algorithmic boundary of 0.10.

Evaluation Metric	Baseline Model (Input Scrubbing Only)	Aligned Model (Adaption Blueprint Applied)	Delta (Δ) Improvement	Status / Interpretation
Demographic Parity (DPD)	0.390	0.014	-0.376	**Passed:** Disparate impact mitigated significantly below the 10% threshold.
Equal Opportunity (EOD)	0.428	0.015	-0.413	**Passed:** True positive approval rates equalized for qualified borrowers.
Model Predictive Accuracy	86.5%	84.8%	-1.7%	**Marginal Trade-off:** Slight accuracy cost yields absolute fairness gains.

When running the data through Adaption's Blueprint constraints, the proxy signatures were successfully decoupled from the risk evaluation. The Aligned system yielded nearly perfect equilibrium, elevating the informal target group's approval rate to **96.6%** compared to the formal group's **98.0%**. The resulting disparities dropped to **0.014 (DPD)** and **0.015 (EOD)**, securing strict structural fairness while preserving the model's core utility.

5. Discussion and Limitations

Broader Safety Implications

Our results clearly illustrate that alternative lending models do not need an explicit identity column to perpetuate historical systemic bias. Through redundant encoding, deep behavioral data traces like transaction patterns act as near-perfect downstream proxies for socio-economic standing. If left unchecked, unaligned fintech models will create an invisible, automated layer of economic exclusion across the Global South.

The marginal 1.7% drop in predictive accuracy observed in our aligned pipeline perfectly reflects the structural Pareto frontier documented by **Kozodoi et al. (2022)**. Their extensive benchmarks of algorithmic credit scoring systems demonstrate that achieving localized demographic equity requires shifting off the absolute global empirical risk minimization peak, rendering minor utility trade-offs both mathematically inevitable and socially necessary.

Limitations

This research sandbox operates on a synthetic distribution of alternative features over a finite test set. In production financial environments, variables are far more chaotic, non-linear, and multi-layered. Furthermore, our current experiment focused strictly on binary credit approval decisions and did not evaluate downstream alignment threats in collection algorithms or algorithmic dynamic interest rate pricing.

Future Work

If selected for the Apart Fellowship, our roadmap focuses on expanding *SutraAudit* from a static sandbox into a dynamic runtime utility:

- **Production SDK:** Developing an open-source CI/CD testing integration tool that financial engineering teams can plug into existing codebases to evaluate dataset distributions automatically before deploying updates.
- **Multilingual Text Auditing:** Expanding the blueprint policy steering to handle unstructured text data, specifically identifying how customer care logs written in regional Indic dialects introduce risk scoring penalties.
- **Causal Graph Structuring:** Creating structural causal graphs to mathematically map out and block proxy leakage vectors across a wider pool of digital footprints.

6. Conclusion

SutraAudit successfully establishes an empirical framework for evaluating and resolving proxy leakage in alternative lending systems, satisfying the core equity requirements of the RBI's "Seven Sutras." By moving the alignment mechanism directly into the data configuration layer using Adaption Labs' platform, we prove that fintech models can achieve deep, structural fairness without sacrificing corporate operational accuracy.

Code and Data

Code repository: `https://github.com/adityatambi/sutraAudit`
Data/Datasets: `https://github.com/adityatambi/sutraAudit/tree/main/data`

LLM Usage Statement

We used Gemini and the Antigravity IDE agent to brainstorm our research approach, generate the localized synthetic dataset, and help draft sections of this report. All experimental results, script executions, and data claims were independently run and verified in our local workspace.

References

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